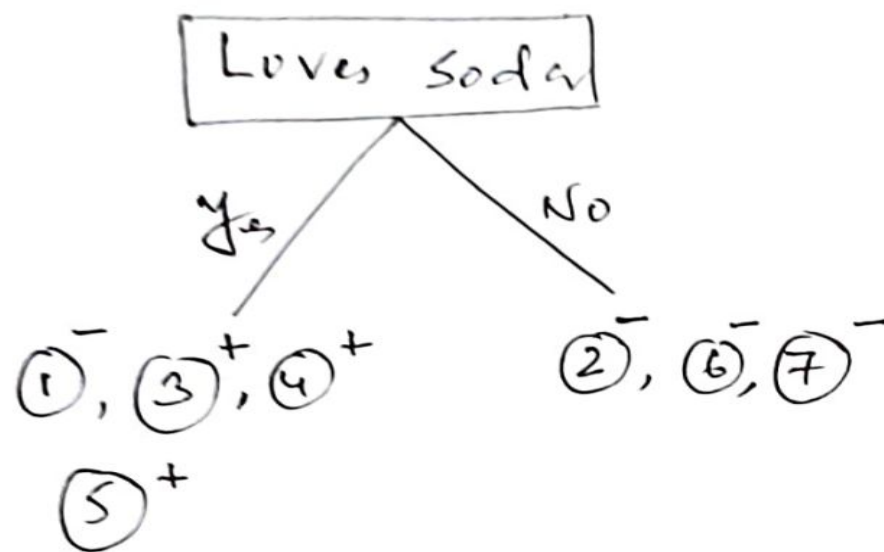
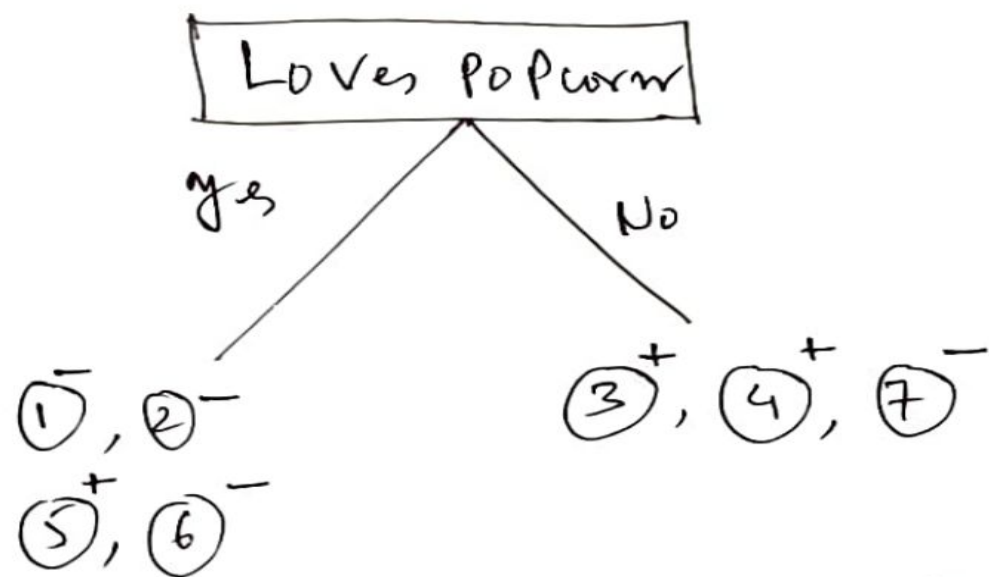


Decision Tree Construction using Information Gain

	Loves Popcorn	Loves Soda	Age	Loves Troll 2
①	Yes	Yes	7	No
②	Yes	No	12	No
③	No	Yes	18	Yes
④	No	Yes	35	Yes
⑤	Yes	Yes	38	Yes
⑥	Yes	No	50	No
⑦	No	No	83	No



$$\text{Info}(\text{Before using Loves Popcorn}) = \text{Info}(3P,4N) = - (3/7) \log(3/7) - (4/7) \log(4/7) = 0.985$$

$$\begin{aligned} \text{Info}(\text{After using Loves Popcorn}) &= (4/7) [-1/4 \log(1/4) - 3/4 \log(3/4)] \\ &\quad + (3/7) [-2/3 \log(2/3) - 1/3 \log(1/3)] \\ &= 0.857 \end{aligned}$$

$$\text{Info Gain}(\text{Loves Popcorn}) = 0.985 - 0.857 = 0.128$$

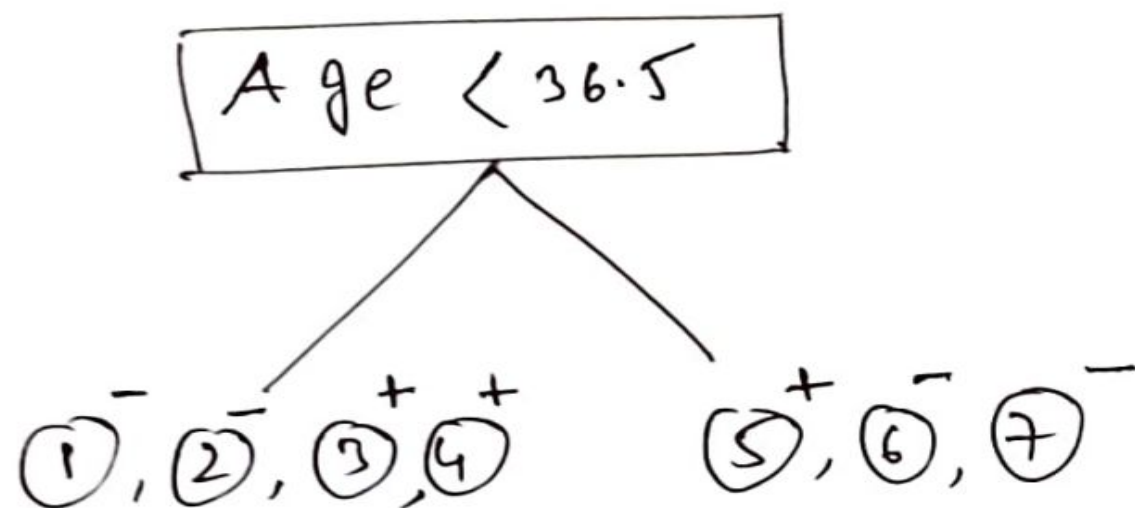
$$\text{Info}(\text{Before using Loves Soda}) = \text{Info}(3P,4N) = - (3/7) \log(3/7) - (4/7) \log(4/7) = 0.985$$

$$\begin{aligned} \text{Info}(\text{After using Loves Soda}) &= (4/7) [-3/4 \log(3/4) - 1/4 \log(1/4)] \\ &\quad + (3/7) [-0/3 \log(0/3) - 3/3 \log(3/3)] \\ &= 0.464 \end{aligned}$$

$$\text{Info Gain}(\text{Loves Soda}) = 0.985 - 0.464 = 0.521$$

Age

7 12 18 35 38 50 83
 ↓ ↓ ↓ ↓ ↓ ↓
 9.5 15 26.5 36.5 44 66.5



$$\text{Info}(\text{Before using Age} < 36.5) = \text{Info}(3P,4N) = - (3/7) \log(3/7) - (4/7) \log(4/7) = 0.985$$

$$\begin{aligned} \text{Info}(\text{After using Age} < 36.5) &= (4/7) [-2/4 \log(2/4) - 2/4 \log(2/4)] \\ &\quad + (3/7) [-1/3 \log(1/3) - 2/3 \log(2/3)] \\ &= 0.965 \end{aligned}$$

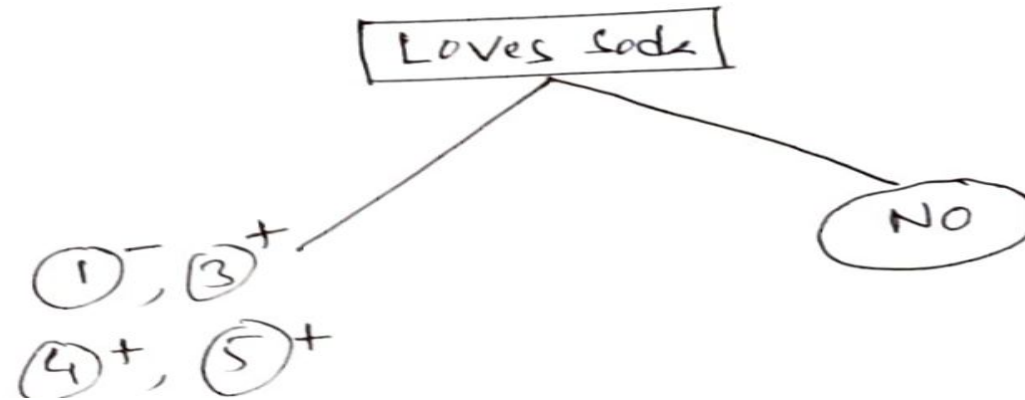
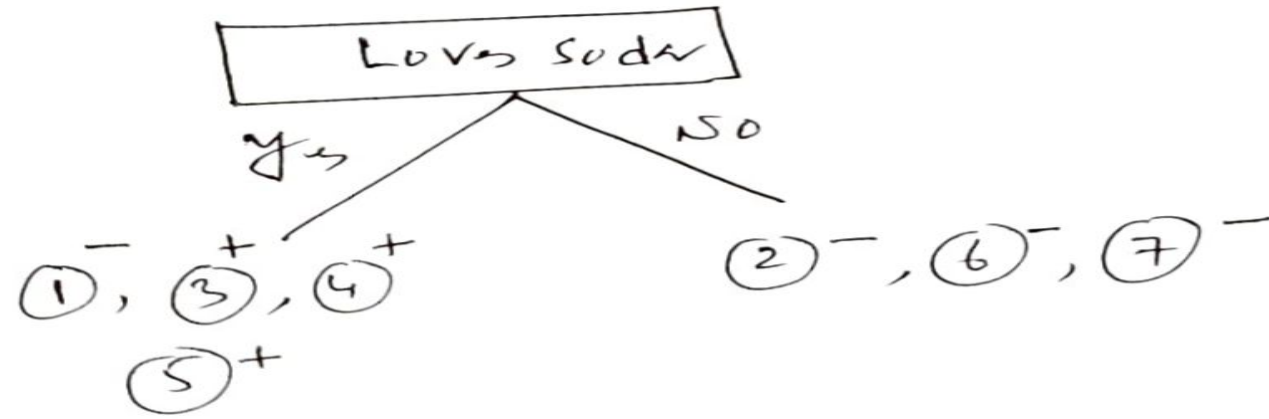
$$\text{Info Gain}(\text{Age} < 36.5) = 0.985 - 0.965 = 0.02$$

We will compute information gain for other midpoints. Assume that the information gain for Age < 15 has got the highest information gain 0.291

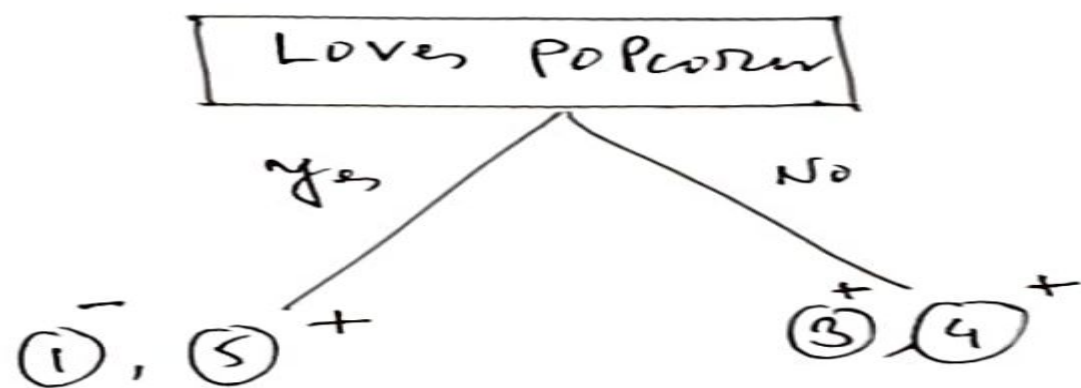
Info Gain(Loves Popcorn) = 0.128

Info Gain (Loves Soda) = 0.521

Info Gain (Age <15) = 0.291



	Loves Popcorn	Loves Soda	Age	Loves Troll 2
①	yes	yes	7	No
③	No	yes	18	yes
④	No	yes	35	yes
⑤	yes	yes	38	yes



$$G(\text{Loves Popcorn}) = 0.25$$

$$\text{Info}(\text{Before using Loves Popcorn}) = \text{Info}(3P,1N) = - (3/4) \log(3/4) - (1/4) \log(1/4) = 0.811$$

$$\text{Info}(\text{After using Loves Popcorn}) = (2/4) [-1/2 \log(1/2) - 1/2 \log(1/2)]$$

$$+ (2/4) [-2/2 \log(2/2) - 0/2 \log(0/2)]$$

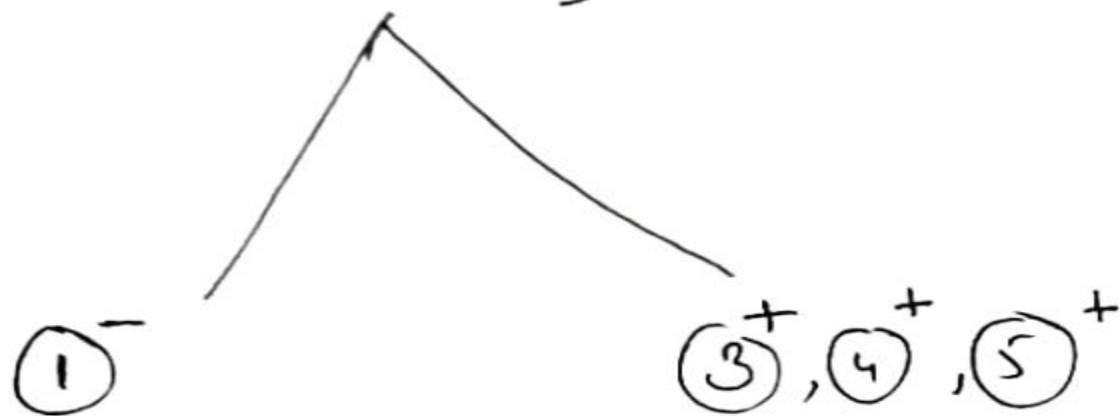
$$= 0.50$$

$$\text{Info Gain}(\text{Loves Popcorn}) = 0.811 - 0.50 = 0.311$$

Age

7	18	35	38
	↓	↓	↓
	12.5	26.5	36.5

~~Age~~ (Age < 12.5)



$$\text{Info}(\text{Before using Age} < 12.5) = \text{Info}(3P, 1N) = - (3/4) \log(3/4) - (1/4) \log(1/4) = 0.811$$

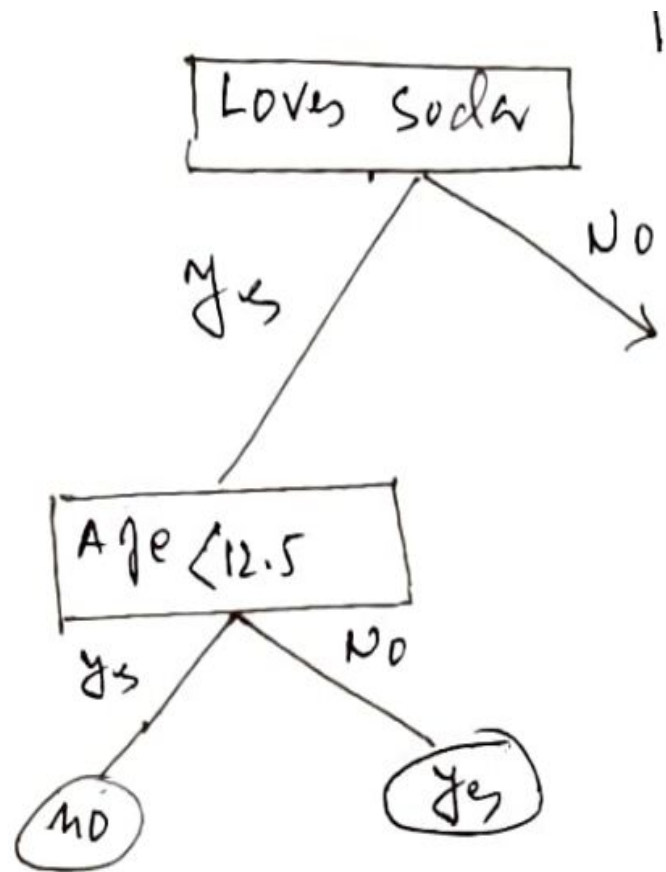
$$\begin{aligned} \text{Info}(\text{After using Age} < 12.5) &= (1/4) [-0/1 \log(0/1) - 1/1 \log(1/1)] \\ &\quad + (3/4) [-3/3 \log(3/3) - 0/3 \log(0/3)] \\ &= 0.00 \end{aligned}$$

$$\text{Info Gain}(\text{Age} < 12.5) = 0.811 - 0.00 = 0.811$$

Info gain of other mid points are less than 0.811

$$\text{Info Gain}(\text{Loves Popcorn}) = 0.811 - 0.50 = 0.311$$

Age < 12.5 is the
winner



To Avoid over fitting we may put a restriction on the number of examples in a ~~leaf~~^{node}. If a ~~leaf~~^{node} contains, i.e., 10 or less examples, this ~~leaf~~^{node} should not be extended further and converted to a leaf. The majority class in the leaf will determine the label of the leaf.

The cross validation approach may be used to find out the appropriate number of examples.

The other more effective approach for avoiding over fitting is tree pruning.